

Arpit Kumar Yadav

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Education

Indian Institute of Technology (ISM) Dhanbad

Expected May 2027

Master in Geotechnical Engineering (CGPA: 9.6)

Dhanbad, Jharkhand

- **Relevant Coursework:** Mechanics of Geomaterials, Advanced Foundation, Unsaturated soil mechanics, Numerical methods in Civil engineering, Ground improvement and Geosynthetics, Computational Geomechanics

Dr. Rammanohar Lohiya Avadh University

July - 2025

Bachelor of Technology Civil Engineering (CGPA: 8.8/10)

Ayodhya, Uttar Pradesh

- **Relevant Coursework:** Construction Technology and Management, Structural Analysis, Soil Mechanics and Foundation Engineering, Surveying, Environmental Engineering, Transportation engineering

Experience

Research Intern

May – July 2024

Indian Institute of Technology Mandi

Mandi, Himachal Pradesh

- Worked on a research project focused on landslide assessment in the Kamand–Mandi region under the guidance of Dr. Dericks P. Shukla
- Collected and analyzed geotechnical and topographical data to evaluate landslide-prone areas and slope conditions.
- Performed laboratory soil characterization tests, including moisture content determination and grain size analysis, to support landslide susceptibility studies.
- Utilized ArcGIS to prepare thematic maps, analyze terrain parameters, and assess landslide vulnerability using spatial data.
- Integrated field observations, soil properties, and GIS-based analysis for landslide hazard assessment and site evaluation.

Projects

Improvement in Static and Dynamic Characteristics of Overburden-Derived Sand through (MICP) Technique | Ongoing

- Conducting research on the application of MICP for sustainable improvement of overburden-derived sand.
- Performed geotechnical characterization of untreated sand through Sieve Analysis, Specific Gravity, Relative Density, and Direct Shear Tests.
- Evaluated particle size distribution, density characteristics, and shear strength parameters to establish baseline soil behavior.
- Preparing MICP-treated specimens to investigate the influence of bio-cementation on the engineering properties of overburden-derived sand.

Performance Evaluation of Self-Curing Concrete Using PEG 400 | Concrete Technology, Sustainable Construction

- Conducted experimental research on the application of Polyethylene Glycol (PEG 400) as a self-curing compound in M20 grade concrete to minimize external water curing requirements.
- Prepared concrete mixes with varying PEG 400 dosages (1%, 1.5%, and 2% by weight of cement) and tested their fresh and hardened properties.
- Evaluated Compressive, Split tensile, and Flexural strengths at 7, 14, and 28 days to compare performance with conventionally cured concrete.
- Found 1% PEG 400 optimal for strength and curing efficiency, reducing water demand while demonstrating self-curing concrete's potential as a sustainable solution in water-scarce regions.

Technical Skills

- **Software & Tools:** AutoCAD, MIDAS, GeoStudio, ArcGIS, MS Excel (Advanced)
- **Programming Computational Skills:** C, Java, MATLAB
- **Laboratory & Field Testing:** Soil Mechanics (CBR, Direct Shear, Triaxial, Consolidation), Concrete Mix Design (IS Codes), Material Testing
- **Engineering Concepts:** Foundation Design, Ground Improvement Techniques, Soil-Structure Interaction, Rock Mechanics, Retaining Structures

Achievements

- * Qualified **GATE 2025 (Civil Engineering)** with [Score-415]
- * Qualified **L&T BIS Examination**, demonstrating strong technical proficiency in civil engineering.
- * Ranked among the **Top 3** students in the B.Tech (Civil Engineering) department for overall academic excellence.
- * Participated in **HACK4SUSTAIN 2025**, a national-level hackathon focused on sustainable and future-ready solutions.

Positions of Responsibility

- **Student Placement Representative (SPR), IIT Dhanbad:** Coordinated between students and the Career Development Centre (CDC), facilitated placement activities, and assisted in organizing recruitment drives.
- **Class Representative (CR) – B.Tech & M.Tech:** Coordinated between faculty and students, managing class communication and academic activities.
- **Member RMLAU Club:** Organized workshops, seminars, events, and hosted club meetings.
- **Volunteer:** Contributed to *Deepotsav, Ayodhya*, supporting large-scale cultural event organization and management.